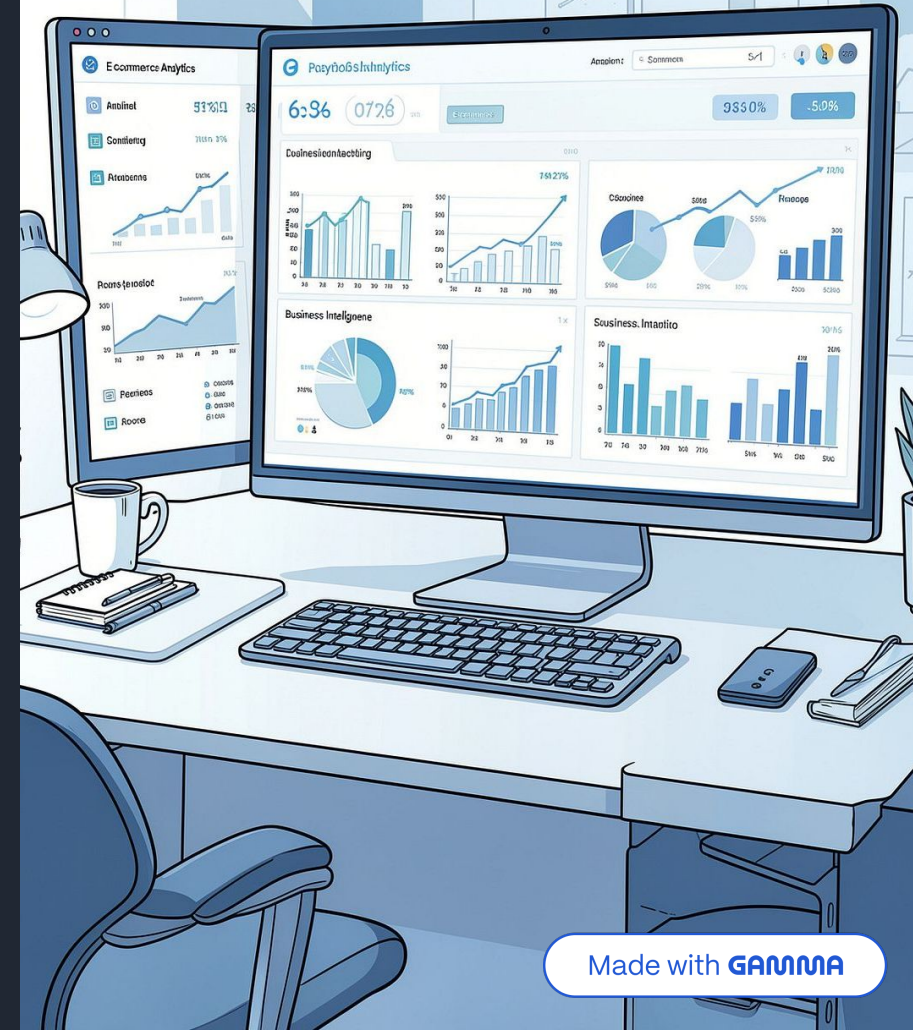


E-Commerce Business Analytics

SQL-Based business analysis using MySQL – verified results from uploaded schema, dataset and KPI queries. This portfolio translates query outputs into actionable business insights for stakeholders.

MYSQL

VERIFIED SQL RESULTS



Business Problem & Objectives



Identify high-value customers

Find top spenders, repeat buyers, and segment-level contributors to revenue.



Product & category performance

Reveal top products, units sold by category, and items above category-average revenue.



Sales channel & AOV

Compare Website, Mobile App, Marketplace revenue and AOV to prioritize channel investments.



Time trends & seasonality

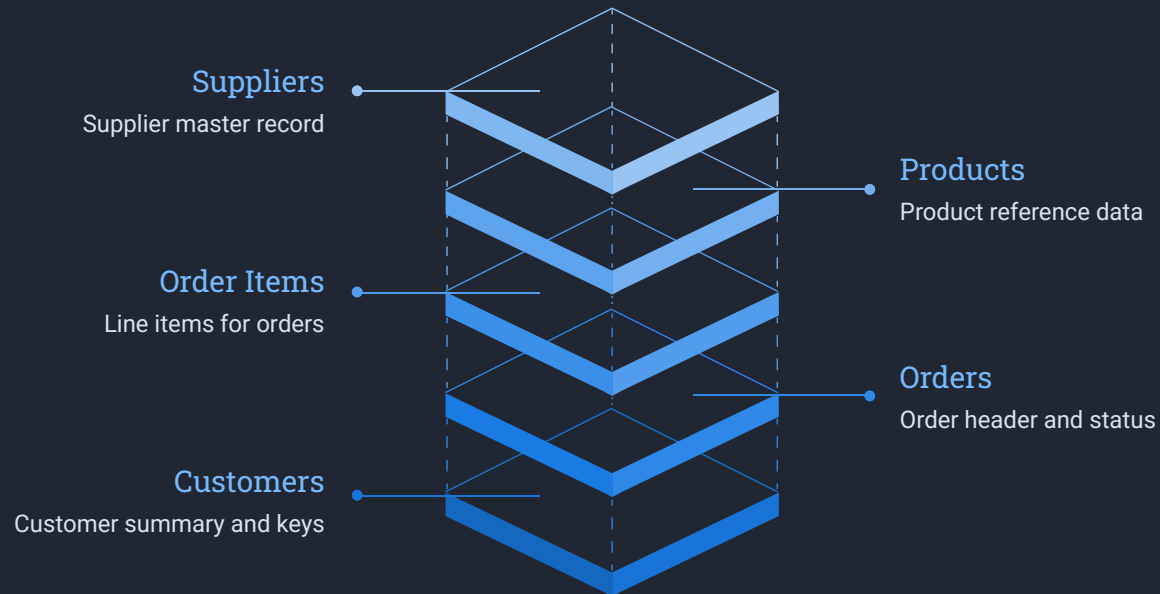
Monthly revenue and MoM growth to spot peaks (e.g., Aug 2025) and troughs (e.g., Jul 2025).



Orders & payments health

Assess delivery rates, cancellations, and payment success by method to reduce friction.

Database & Data Model



Key tables & relationships

- Customers — Orders — Order Items (transactional backbone)
- Products — Categories — Suppliers (product attributes & taxonomy)
- Payments (payment method, success rates) and Reviews (ratings)
- Addresses (delivery performance analysis)

Transactional joins between orders, order_items and payments enable revenue, AOV, channel and operational analysis.

Executive Business KPIs

Total Orders

5,000

Total Units Sold

22,440

Total Revenue

₹281,609,068.11

Average Order Value (AOV)

₹56,321.81

Total Customers

1,000

Total Products

300

Delivered Orders

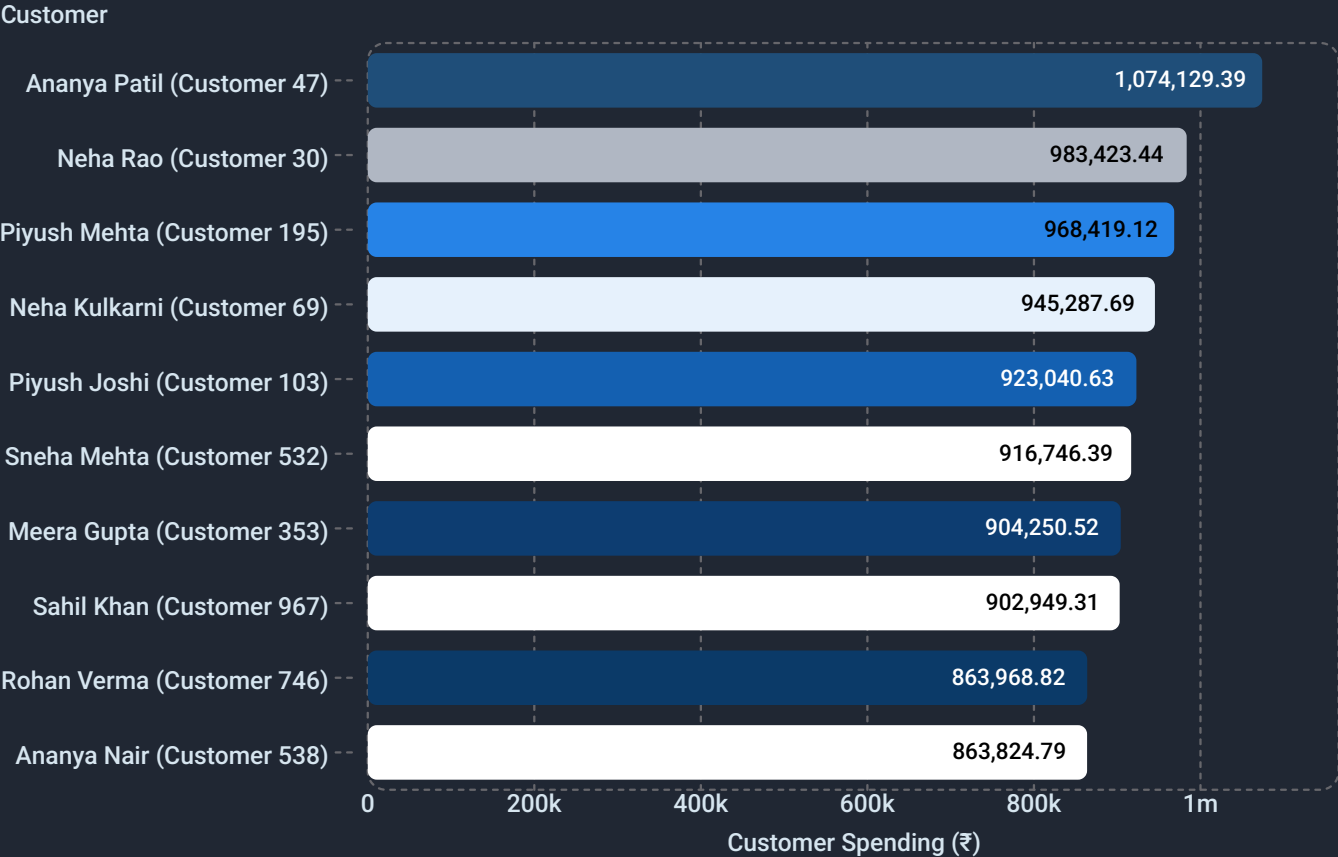
3,394 (67.88%)

Cancelled Orders

414 (8.28%)

Observations: Delivered orders account for ~68% of order volume; Regular customers contribute the largest revenue share (see next slide).

Customer Insights



- Top Customer

Ananya Patil (customer 47):
₹1,074,129.39 (highest spend).

- Average Spend

Average customer spending:
₹284,166.57 – 421 customers exceed this.

- Repeat & Multi-channel Behavior

892 customers placed ≥ 3 orders; 888 customers used multiple sales channels.

- High Engagement Cohorts

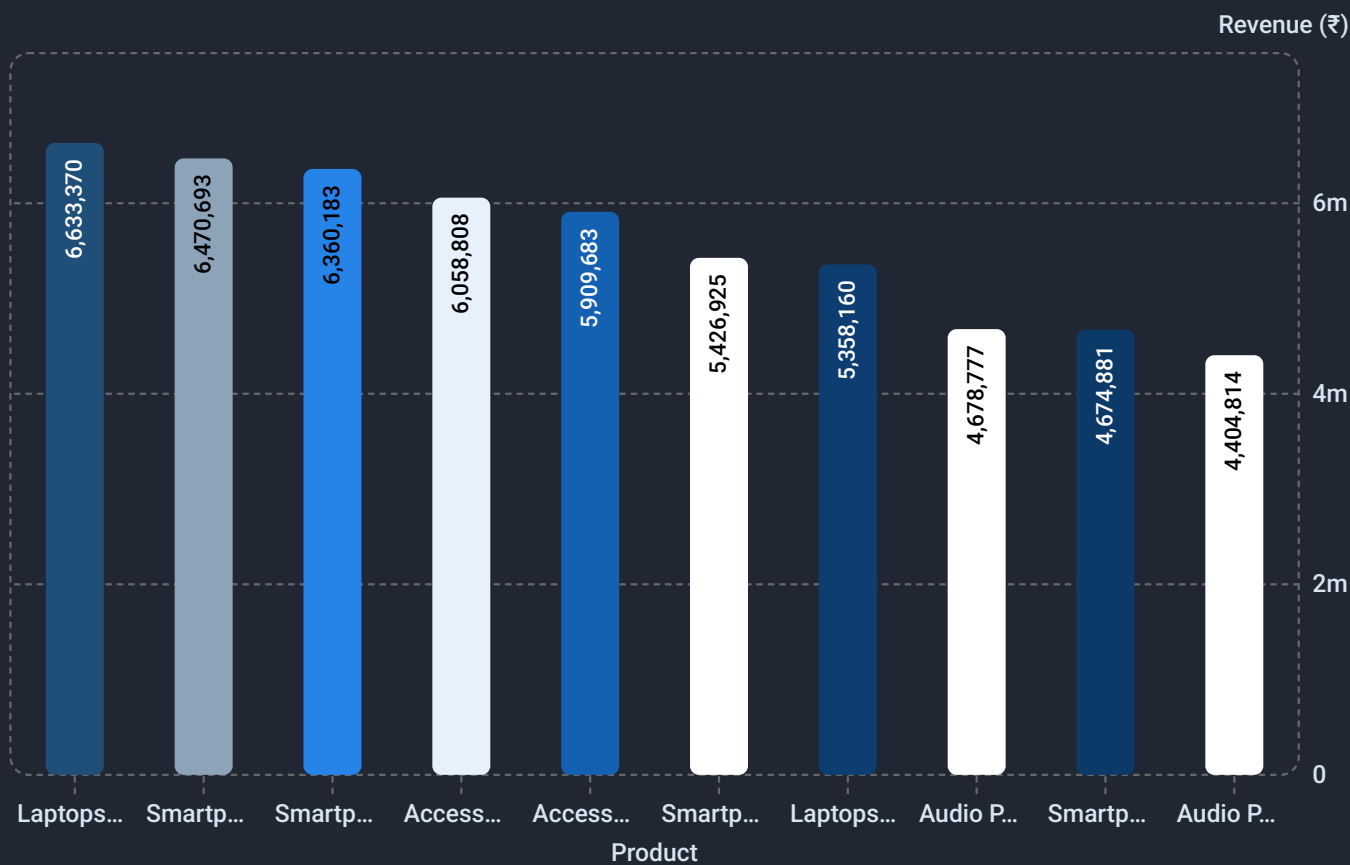
872 customers placed ≥ 3 orders and spent $>₹50,000$; 965 customers bought from ≥ 3 categories.

- Orders Leader

Aditi Jain placed the most orders: 14.

📄 All figures are derived from uploaded SQL results (e.g., Q1, Q2, Q3, Q27).

Product & Category Insights



Top 10 Products by Revenue



Top Product

Laptops Product 017:
₹6,633,370.47 (top revenue-generating product).



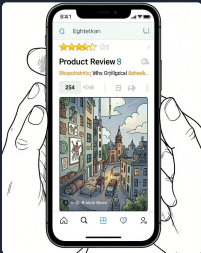
Category Revenue

Electronics:
₹118,230,252.79 (41.98%);
Home: ₹74,867,057.67 (26.59%).



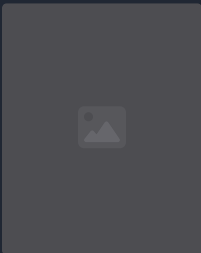
Units Sold

Grocery leads units sold among top-5 categories: 2,998 units; Beauty 2,932; Home 2,891.



Quality & Revenue

61 products have above-average revenue and an average rating ≥ 4.0 (Q40).

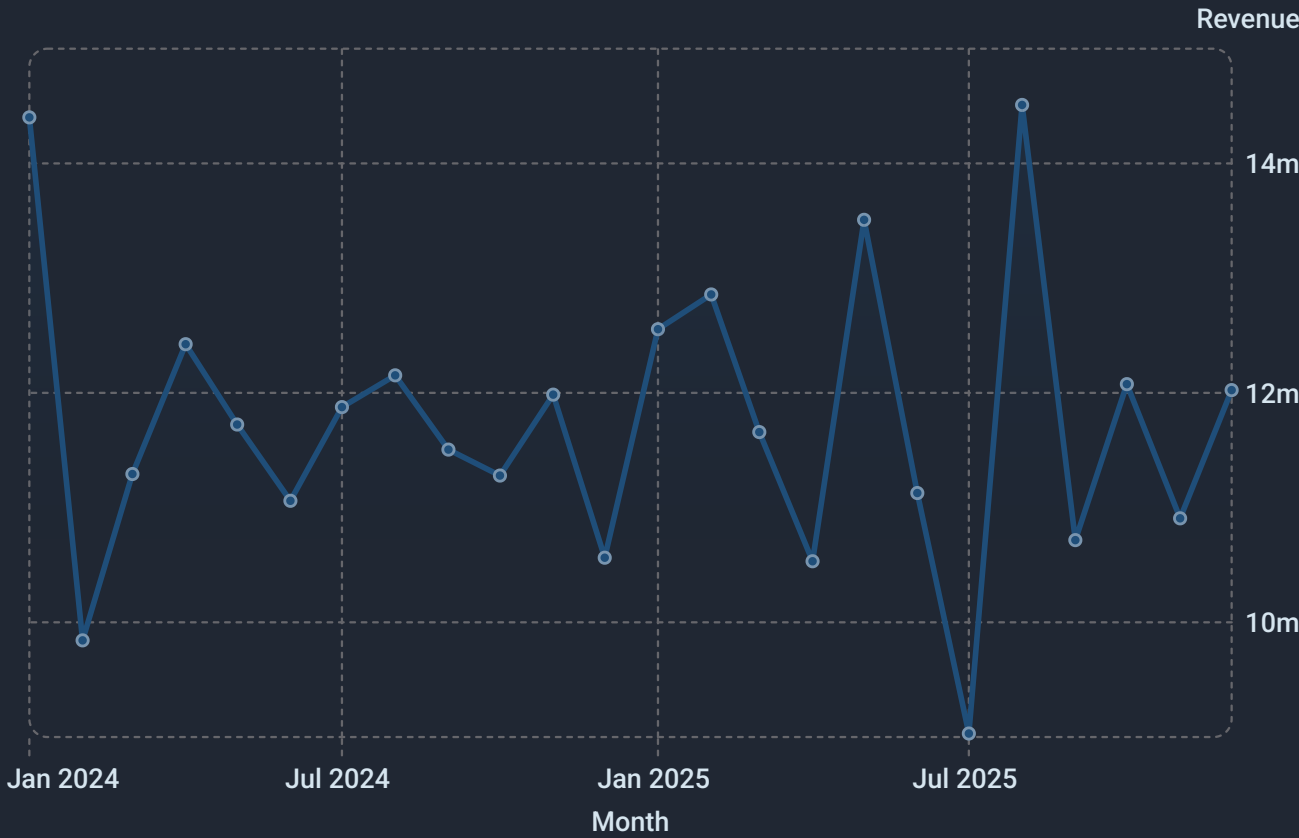


Category Benchmark

148 products have revenue above their category average.

Sales Channels & Time Analysis

Monthly Revenue Trend — Jan 2024 to Dec 2025



Channel Revenue

Website: ₹125,578,337.74 (44.59%) •
Mobile App: ₹109,612,180.45 (38.92%) •
Marketplace: ₹46,418,549.92 (16.48%).

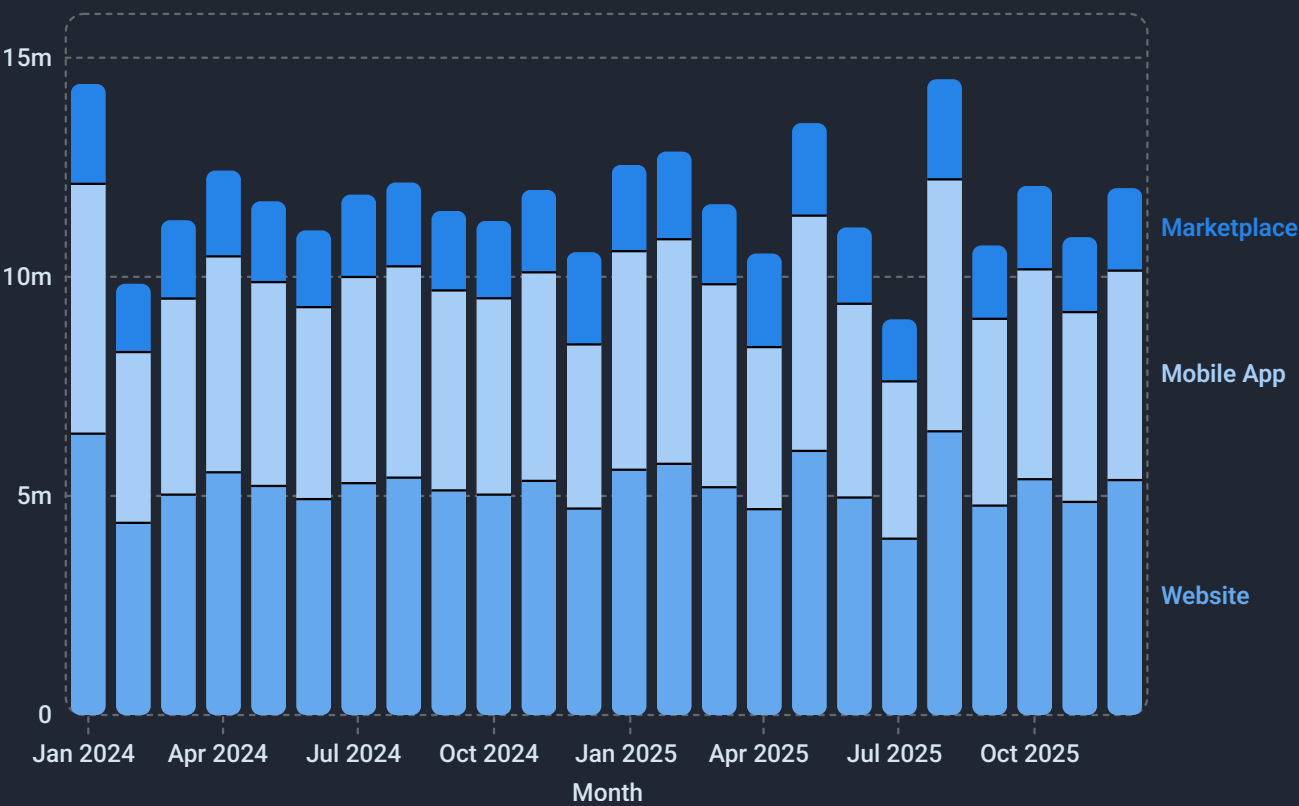
AOV by Channel

Marketplace ₹56,885.48 • Mobile App
₹56,793.88 • Website ₹55,713.55 (Q12).

Category × Channel

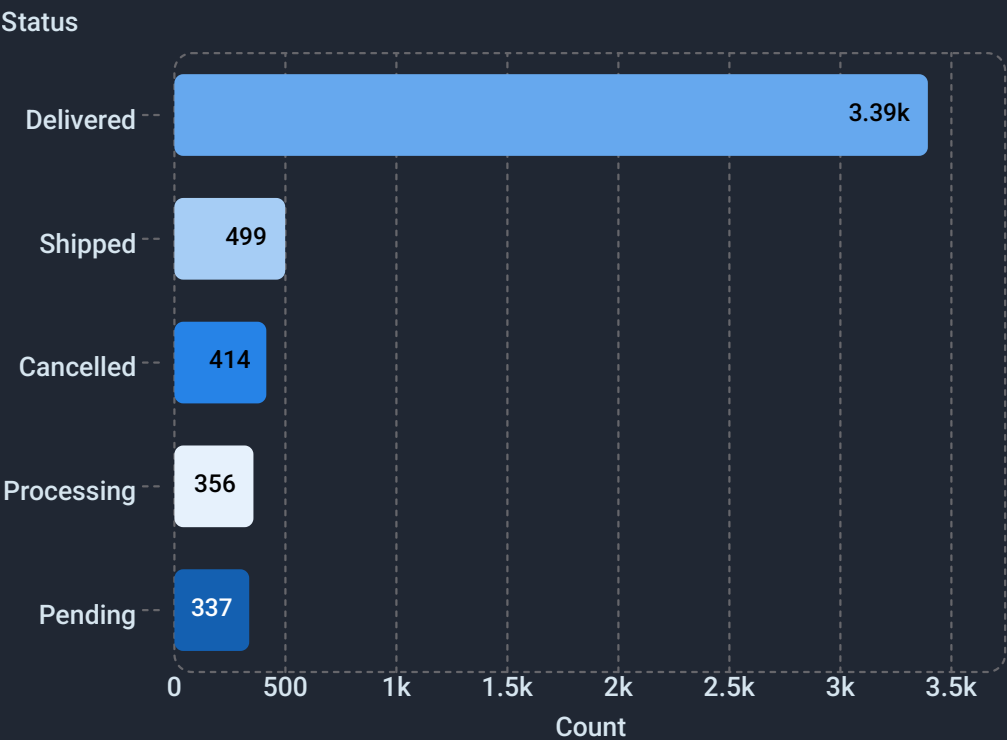
Electronics is top in every channel
(Website ₹52,875,745.66; Mobile App
₹44,129,259.81; Marketplace
₹21,225,247.32).

Monthly Revenue by Sales Channel



Time highlights: August 2025 highest monthly revenue ₹14,509,869.24 (240 orders); July 2025 lowest ₹9,030,706.54 (182 orders). December 2025: ₹12,024,279.72 with 10.25% MoM growth vs November 2025. February 2024: -31.66% MoM vs January 2024 (Q6, Q14, Q41).

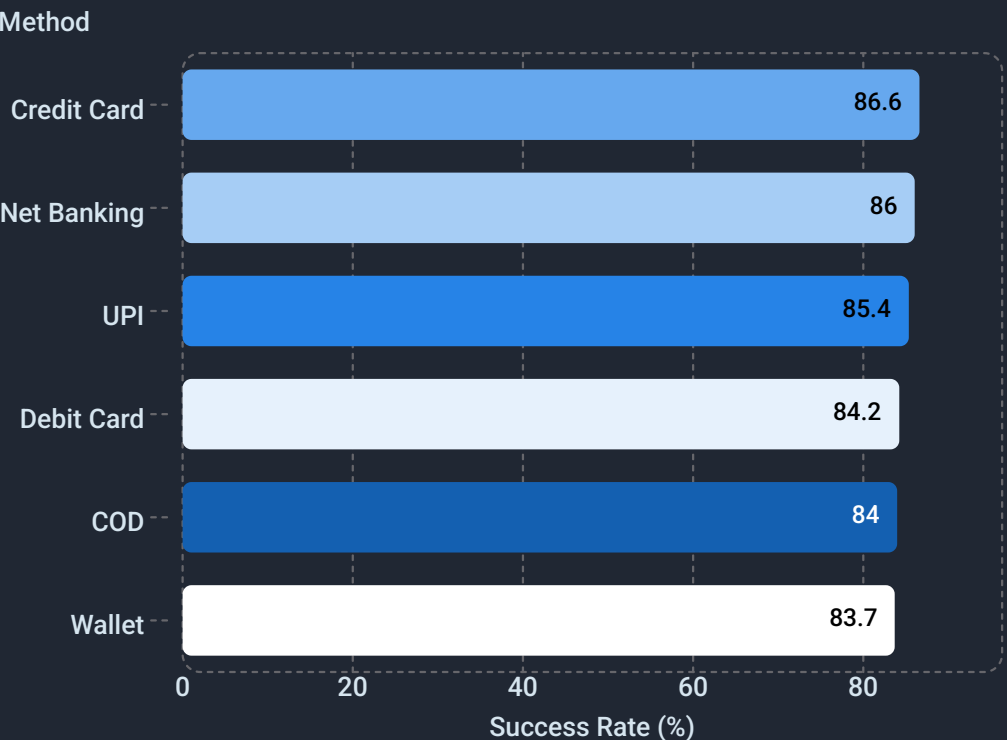
Orders, Payments & Operations



Order Count by Status

- Order Status Revenue

Delivered: ₹191,583,331.24 • Shipped: ₹27,315,086.72 • Cancelled: ₹25,455,001.99 • Processing: ₹19,668,721.43 • Pending: ₹17,586,926.73 (Q16).



Payment Success Rate by Method

- Order Counts & Rates

Delivered 3,394 (67.88%), Shipped 499 (9.98%), Cancelled 414 (8.28%), Processing 356 (7.12%), Pending 337 (6.74%).

- Payment Success Rates

Credit Card 86.60% • Net Banking 86.04% • UPI 85.35% • Debit Card 84.23% • COD 83.98% • Wallet 83.69% (Q20).

Operational focus: reduce cancellations and improve payment success on lower-performing methods to lift revenue and delivery yield.

Advanced SQL Findings (Analytics Methods)

Customer Lifetime & Ordering Patterns

First vs latest order dates and days between (Q33); running sums of customer spending using window functions (Q34).

Ranking & Cohorts

Top orders per customer via RANK() (Q35); customer spend ordered with LAG() comparisons (Q31).

Product & Repeat Purchase Analytics

Customer-product repeat purchases (232 combinations, Q38); product ranking within categories and identification of 148 products above category-average revenue (Q22, Q32).

Channel & Time Series Calculations

Monthly revenue and MoM growth for each channel across 24 months (Q41, Q42).

These advanced queries enable defensible metrics and cohort-level diagnostics — each insight maps to a numbered SQL query in the uploaded project (refer to the Interview-Ready Story).

Recommendations & Next Steps

1. Prioritize Electronics & High-ROI SKUs

Electronics contributes 41.98% of revenue; focus promotions and inventory on top products (e.g., Laptops Product 017).

2. Improve Delivery Yield & Reduce Cancellations

Address causes behind 8.28% cancellations and lift delivered share above 67.88%.

3. Evaluate Website and Mobile App Performance Further

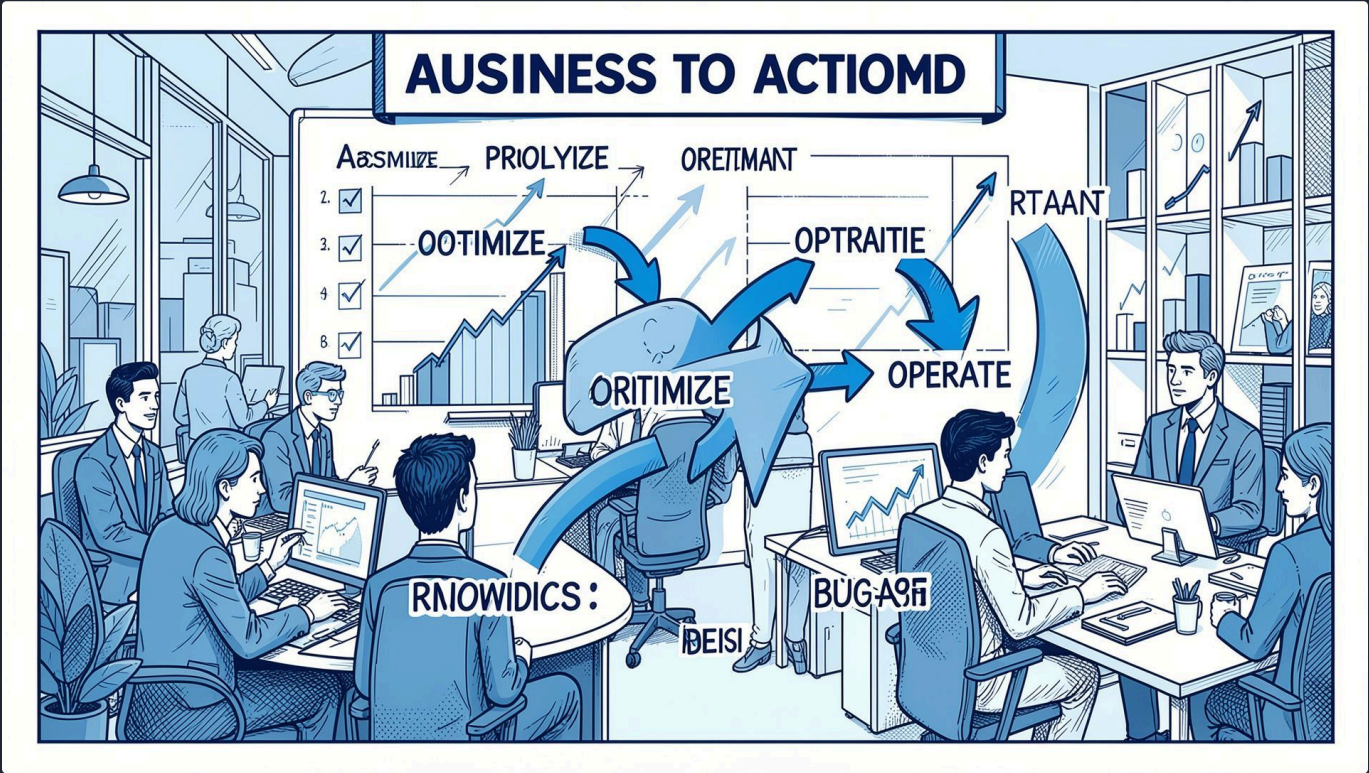
Use conversion, acquisition cost, and retention metrics before reallocating channel investment. Current AOVs are similar across channels (₹55,713–₹56,885), suggesting channel choice may be driven by customer preference rather than value.

4. Loyalty & Cross-Category Growth

Target customers who buy across ≥ 3 categories and those above-average spenders (421+ customers) with retention offers.

5. Operational & Payment Improvements

Raise success rates for lower-performing payment methods and streamline fulfillment to protect revenue.



❏ Validation: Every numerical insight here links back to a specific query in the uploaded project (e.g., Q7, Q9, Q16, Q20, Q40).